

VAMSHI KARANAM

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Professional Summary

Assistant Professor of Geology at the University of Arkansas at Little Rock, specializing in InSAR, GNSS, and GIS to investigate land subsidence, groundwater- and fluid-driven deformation, critical-zone processes, and environmental and infrastructure hazards. My research integrates satellite geodesy, physics-based poroelastic modeling, and geospatial analytics to study human-environment interactions in sedimentary and urban systems. I teach and mentor students in remote sensing, environmental science, and GIS while establishing a computational geophysics and remote sensing research program at UALR.

Experience

Assistant Professor of Geology, University of Arkansas at Little Rock 2026 – Present

- Teach Earth and the Environment, Earth and the Environment Lab, and GIS I.
- Continuing research on InSAR- and GNSS-based deformation monitoring and poroelastic modeling of environmental and energy-related hazards.

Research Assistant, SMU Radar Lab, Dallas, USA 08/2021 – 12/2025

Advisor: Dr. Zhong Lu, Professor

- Produced long-term InSAR/GNSS deformation products to assess land subsidence, strain, and infrastructure risk in the Permian Basin; investigated wastewater-related well blowouts using multi-sensor remote sensing and subsurface modeling.
- Developed geocoded SLC offset tracking and 3D deformation workflows for the NISAR mission, and 3D poroelastic models to quantify subsurface stress evolution.
- Led collaborations on seismicity–subsidence coupling and infrastructure exposure assessment; led a two-day Google Earth Engine workshop introducing students to large-scale geospatial analysis.

Research Assistant, Dept. of Geospatial Engineering, IIT Roorkee, India 02/2021– 06/2021

Advisor: Dr. Saurabh Vijay, Asst. Professor

- Modeled flooding–subsidence interactions using machine-learning and GIS workflows to assess environmental risk in urban systems.

Research Intern, German Research Center for Geosciences (GFZ), Germany 03/2020 – 06/2020

Advisor: Dr. Mahdi Motagh, Professor

- Mapped groundwater-driven subsidence in Delhi using InSAR and in-situ datasets, delineating deformation hotspots with implications for urban safety and water management.

Visiting Researcher, Leibniz University Hannover (LUH), Germany 09/2019 – 02/2020

Advisor: Dr. Mahdi Motagh, Professor

- Quantified coal fire–induced subsidence across a large mining region in India by integrating thermal anomaly mapping using Landsat-8 with InSAR deformation products.
- Teaching: instructed/assisted courses in remote sensing, GIS, radar remote sensing, earth systems, and photogrammetry across SMU, IIT Roorkee, and LUH;
- designed hands-on labs in InSAR processing and field GNSS/total-station surveying.

Grants & Awards

- **Geospatial World Rising Star Award**, Geospatial World Forum, Netherlands 2026
- **NASA FINESST Research Grant**, \$88,700, SMU, USA 2024 – 2026
- **DAAD KOSPIE Fellowship**, \$11,000, Leibniz University Hannover, Germany 2019 – 2020
- **Gold Medal**, M.Tech, Geomatics Engineering, IIT Roorkee, India 2020
- **MHRD National Graduate Scholarship**, Government of India 2018 – 2020
- **Professional Development Microgrants**, IEEE GRSS 2023
- **IEEE IGARSS Travel Grant**, IEEE GRSS 2023

Technical Skills

- **Programming:** Python, MATLAB, Bash, Google Earth Engine (JavaScript)
- **Remote Sensing:** ERDAS Imagine, GAMMA, StaMPS, MintPy, SARscape, ISCE, SNAP
- **GIS & Modeling:** ArcGIS Pro, QGIS, AutoCAD, COMSOL Multiphysics, GBIS
- **Remote Sensing Analytics:** InSAR time-series analysis, 3D deformation retrieval, poroelastic modeling, geospatial data fusion, change detection, hazard mapping

Education

Ph.D. in Geophysics (*Environmental & Applied Geoscience*) 2021 – 2026

Southern Methodist University (SMU), Dallas, USA, (GPA 3.98)

Dissertation: *Satellite-based Deformation Monitoring and Poroelastic Modeling of Environmental and Energy-Related Geohazards in the Permian Basin*

M.Tech in Geomatics Engineering 2018 – 2020

Indian Institute of Technology (IIT) Roorkee, India, (Gold Medal, GPA 3.9)

Dissertation: *Geospatial modeling of coal fires on mine subsidence in Jharia Coalfields, India*

Bachelor of Architecture 2013 – 2018

National Institute of Technology (NIT) Calicut, India

Publications

Journal Articles:

- **Karanam, V.**, Lu, Z., Kim, J., (2024). Investigation of Oil Well Blowouts Triggered by Wastewater Injection in the Permian Basin, USA. *Geophysical Research Letters* 51, e2024GL109435. <https://doi.org/10.1029/2024GL109435>.
- **Karanam, V.**, Lu, Z., (2023). Hydrocarbon production induced land deformation over Permian Basin; analysis using persistent scatterer interferometry and numerical modeling. *International Journal of Applied Earth Observation and Geoinformation* 122, 103424. <https://doi.org/10.1016/j.jag.2023.103424>.
- **Karanam, V.**, Motagh, M., Garg, S., & Jain, K. (2021). Multi-sensor remote sensing analysis of coal fire induced land subsidence in Jharia Coalfields, Jharkhand, India. *International Journal of Applied Earth Observation and Geoinformation*, 102, 102439. <https://doi.org/10.1016/j.jag.2021.102439>.
- Sui, Q., Lu, Z., Meng, T., Kim, J.-W., Higman, B., Dai, C., Budukumah, E., McColl, S., Howat, I., Hults, C., **Karanam, V.**, Liang, K. Hydrometeorological and Topographic Controls Govern

Spatiotemporal Heterogeneity in Rock Glacier Kinematics: Insights from Satellite Interferometry in Southeastern Alaska, *Journal of Geophysical Research: Earth Surface*, 131(4).
<https://doi.org/10.1029/2025JF008895>.

- Aziz Zanjani, A., DeShon, H.R., **Karanam, V.**, Savvaidis, A., (2024). Insights into Temporal Evolution of Induced Earthquakes in the Southern Delaware Basin Using Calibrated Relocations from the TexNet Catalog (2017–2022). *Earth and Space Science*, 12, e2024EA004027.
<https://doi.org/10.1029/2024EA004027>.
- Liang, K., Kim, J., Lu, Z., Fattahi, H., Bato, M.G., Brancato, V., Jeong, S., **Karanam, V.** (2025). Offset tracking with geocoded SLC. *IEEE Transactions on Geoscience and Remote Sensing*.
<https://doi.org/10.1109/TGRS.2025.3570627>.
- Aziz Zanjani, A., DeShon, H.R., Karanam, V., Savvaidis, A., (2024). Insights into Temporal Evolution of Induced Earthquakes in the Southern Delaware Basin Using Calibrated Relocations from the TXAR Catalog (2009–2016). *The Seismic Record* 4, 140–150. <https://doi.org/10.1785/0320240011>.
- Garg, S., Motagh, M., Indu, J., & **Karanam, V.** (2022). Tracking hidden crisis in India’s capital from space: Implications of unsustainable groundwater use. *Scientific Reports*, 12(1), 651.
<https://doi.org/10.1038/s41598-021-04193-9>.

In Preparation/Under Review

- **Karanam, V.**, Lu, Z. An Integrated Workflow for Quantifying Fluid-Induced Deformation Using InSAR and 3D Poroelastic Modeling.
- **Karanam, V.**, Lu, Z., Kim, J. Energy Infrastructure Monitoring with InSAR in Central Basin Platform, West Texas, USA.

Selected Conference Presentations:

Oral Presentations

- **Karanam, V.**, Lu, Z., (2025). Poroelastic Modeling and InSAR Analysis of Hydrocarbon Production-Induced Surface Deformation in the Permian Basin, USA. *EGU General Assembly Conference Abstracts*, EGU25-14878.
- **Karanam, V.**, Lu, Z. and Kim, J. (2023) Geophysical Characterization of Oil Well Blowouts Triggered by Pore Pressure Propagation from Wastewater Injection Through Hydrogeologic Structures, In *AGU Fall Meeting 2023 Abstracts*.
- **Karanam, V.**, Lu, Z. and Kim, J.-W. (2023). Hydrocarbon Production Induced Land Deformation Over Delaware Basin, Analysed Using Persistent Scatterer Interferometry. *2023 IEEE International Geoscience and Remote Sensing Symposium*, Pasadena, CA, USA: IEEE, pp. 1846–1849.
- Lu, Z., Zheng, W., **Karanam, V.**, Kim, J., (2023). Human-induced geohazards in Permian Basin, USA revealed by InSAR and numerical modeling. *EGU General Assembly Conference Abstracts*, EGU23-2152

Poster Presentations

- **Karanam, V.**, Motagh, M., Garg, S., & Jain, K. (2021). Combined Effect of Mining, Subsidence and Coal Fires in Jharkhand, India Investigated using Satellite Remote Sensing and Data Fusion. In *AGU Fall Meeting 2021 Abstracts* (Vol. 2021, pp. NH15D-0481).
- Aziz Zanjani, A., DeShon, H., **Karanam, V.**, Savvaidis, A., (2024), Spatiotemporal Evolution of Induced Earthquakes in the Southern Delaware Basin, Reeves-Pecos, West Texas, *SSA Annual Meeting 2024 Abstracts*.

- Garg, S., **Karanam, V.**, Motagh, M., Mishra, V., Xia, Z., Shevchenko, A. V., Stefanova Vassileva, M., Roessner, S. (2023). Monitoring land subsidence in Joshimath, Uttarakhand using InSAR: A preliminary study, XXVIII General Assembly of the International Union of Geodesy and Geophysics, Berlin.
- Garg, S., **Karanam, V.**, & Motagh, M. (2021). The continuous sinking of National Capital Region, India—Investigated using the Sentinel-1 time series InSAR approach. *FRINGE Workshop* 2021.

Professional Service

- **Manuscript Reviewer**
 - IEEE Transactions on Geoscience and Remote Sensing, Engineering Geology, Advances in Space Research, Natural Hazards & Risk, Science of the Total Environment, Remote Sensing, etc.
- **Organizing Committee**, International Conference on Unmanned Aerial Systems in Geomatics 2020
- **President**, Society of Exploration Geophysicists (SEG) SMU Student Chapter (2025)
- **Member**, AGU, EGU, IEEE & AAPG
- **Registered Architect**, Council of Architecture, India

Selected Media Coverage

- **Bloomberg (2025):** "[Texas oil boom spawns a toxic crisis of the Industry's own making](#)"
- **The Dallas Morning News (2024):** "[Is it natural or fracking? SMU study reveals cause of past Texas earthquakes](#)"
- **Houston Chronicle (2024):** "[Study: More blowouts to come](#)"
- **The Texas Tribune (2024):** "[Ranchers reported abandoned oil wells spewing wastewater](#)"
- **Newsweek (2024):** "[Well Blowouts in Texas Prompt \\$100M Emergency Funding Plea](#)"
- **Carlsbad Current-Argus (2023):** "[Oil and gas is 'deforming' New Mexico's land, study says](#)"
- **BBC (2023):** "[How a Delhi district stopped the ground from sinking](#)"
- **Hindustan Times (2022):** "[Excessive groundwater extraction causing parts of Delhi-NCR to sink](#)"